



VoxSentinalX — SIH 2026 Demo Video Script

Problem Statement #26104 | Real-Time AI Voice Deepfake Detection

Target Duration: 2:30 – 2:50



Pre-Recording Setup Checklist

Before pressing record, have everything ready:

#	ITEM	DETAILS
1	Backend running	<code>run_all.bat</code> or manually: <code>cd backend && python run.py</code>
2	Frontend running	Should auto-open at <code>http://localhost:3000</code>
3	Browser	Chrome fullscreen (F11), 90% zoom, dark mode
4	Screen resolution	1920×1080 (16:9)
5	Screen recorder	OBS Studio recommended (1080p, 60fps)
6	Mic	Headset mic preferred (avoids feedback during speakerphone demo)
7	Phone ready	Load a deepfake audio sample (e.g., AI-cloned Elon Musk from <code>dataset_samples/</code>)
8	Real voice	You'll speak naturally into the mic as contrast
9	Tabs open	Tab 1: Landing page (<code>localhost:3000</code>), nothing else visible
10	Notifications	Close all — no popups, no Discord, no Windows notifications



SCENE 1 — The Hook

🕒 0:00 – 0:20 (20 seconds)

[SCREEN: Showcase Landing Page — hero section with animated waveform visualizer]



VOICEOVER (speak naturally, slightly urgent):

"Imagine getting a phone call from your bank manager — the voice sounds exactly like them, the tone, the accent, everything matches. But it's not them. It's an AI-generated deepfake clone, and they're about to steal your life savings."

🔴 **What to do on screen:** - Let the landing page load with the animated waveform hero visualizer playing - The VoxSentinalX logo and tagline are visible - Slowly scroll down to reveal the **10-Vector Forensic Bento Grid**



VOICEOVER:

"This is the problem we're solving. Meet VoxSentinalX — a real-time AI voice deepfake detection platform built for SIH 2026."



SCENE 2 — The 10-Layer Engine

🕒 0:20 – 0:50 (30 seconds)

[SCREEN: Still on Landing Page — Forensic Bento Grid]



VOICEOVER (confident, educational):

"Unlike traditional detectors that rely on a single model, VoxSentinalX runs 10 forensic detection layers simultaneously."

🔴 **What to do on screen:** - Hover over / click through 4 key bento cards:

CARD TO HIGHLIGHT	WHAT TO SAY
Spectral Flatness & Phase	"We check if the frequency spectrum is unnaturally flat — something AI voices always get wrong."
Breathing Cadence	"Real humans breathe. AI doesn't. We detect the absence of natural respiratory pauses."
Jitter & Shimmer	"Human vocal cords have micro-imperfections — tiny pitch wobbles. AI voices are too perfect."
Deep Neural LCNN	"And our neural network — an LCNN with BiLSTM and attention — learns patterns no handcrafted feature can catch."

 VOICEOVER:

"Each layer produces a score. A weighted fusion engine combines them into one real-time risk verdict."

✦ **Then quickly:** - Scroll to the **Threat Simulation Sandbox** - Click **"Wire Fraud Clone"** → radial risk gauge animates to HIGH (red) - Click **"Authentic Human Caller"** → gauge drops to LOW (green)

 SCENE 3 — Live Call Detection Demo ★

🕒 0:50 – 1:40 (50 seconds) — THE MONEY SHOT

[SCREEN: Click **"Live Monitor"** tab in the navbar]

 VOICEOVER (building energy):

"But let me show you the real magic — live, real-time detection."

✦ **Step-by-step actions:**

STEP	ACTION	WHAT TO SAY
1	Click "Live Call Monitor" in navbar	—
2	Dashboard opens — oscilloscope flat, radar empty	—
3	—	"Here's the scenario: I receive a suspicious phone call. I put my phone on speaker next to my laptop mic."
4	Click "Start Live Monitoring" → Allow mic	—
5	Oscilloscope starts showing ambient noise	—
6	 Play deepfake audio from your phone speaker	"I'm now playing an AI-cloned voice through my phone speaker — watch what happens."
7	 THE WOW MOMENT — watch the dashboard light up:	<i>(Pause 2-3 sec. Let the visuals speak.)</i>
• Oscilloscope shows waveform		
• 10-axis radar expands with LOW breathing/jitter		
• Risk gauge climbs green → yellow → RED		
• Diagnostic feed shows alerts		
• Emergency overlay may trigger at >85%		
8	—	"Instantly — spectral flatness flagged, breathing patterns absent, jitter too stable. The system classified this as a deepfake with 92% confidence. And it shows exactly WHY — not just a black-box label."

STEP	ACTION	WHAT TO SAY
9	Stop deepfake. Speak into mic yourself (few sentences)	"Now listen to a real human voice — mine."
10	Radar fills out, gauge drops to green	"Full breathing cadence, natural pitch variance, organic jitter. Verified human. That's the difference."
11	Click "Stop Monitoring"	—



SCENE 4 — File Upload Analysis



1:40 – 2:10 (30 seconds)

[SCREEN: Click "File Analyzer" tab]



VOICEOVER (fast but clear):

"We also support offline forensic analysis. Drag and drop any audio or video file."



What to do on screen: 1. Click **"Audio & Video Analyzer"** tab 2. Drag dataset_samples/Elon Musk.wav into the upload zone 3. Analysis runs — loading animation plays 4. Results appear: forensic timeline, 10-vector breakdown, risk peaks, diagnostic report



VOICEOVER:

"In under 3 seconds, we get a complete forensic timeline — every second scored across all 10 vectors. Risk peaks are highlighted, and actionable countermeasures are generated automatically."



SCENE 5 — Training & Benchmarks + Close



2:10 – 2:40 (30 seconds)

[SCREEN: Click "LCNN Training" tab]

 VOICEOVER (proud, closing):

"Our model is trained on a unified corpus of over 5,400 audio files — covering OpenAI, Coqui XTTS, Seed-TTS, VALL-E, ASVspoof, and Indic language generators."

✦ **What to do on screen:** 1. Click **"LCNN Training"** tab 2. Show metrics: accuracy 86.27%, EER 6.86%, F1 87.63% 3. Scroll to show benchmark comparison chart

 VOICEOVER:

"86% accuracy. 6.8% equal error rate. Tested against 11 different voice cloning engines. And it runs in real-time — no cloud dependency, no GPU required."

✦ **Final shot:** - Scroll back to landing page hero - Hold on the VoxSentinalX logo for 3 seconds

 VOICEOVER (mic drop energy):

"VoxSentinalX — because in a world where any voice can be faked, trust needs to be verified. Thank you."

[Fade to black]

 Pro Tips for Natural Delivery

1. **Don't read word-for-word** — internalize the key points and speak naturally
2. **Pause after the detection moment** — let the visual do the talking for 2–3 seconds before narrating
3. **Show genuine reaction** — a subtle "Whoa, look at that" or "There it is" when the gauge goes red feels authentic
4. **Vary your pace** — slow down for the hook and detection moment, speed up for tech specs
5. **Use your cursor as a pointer** — hover over key UI elements as you explain them
6. **Record in one take** — cuts in a 2-min video feel jarring
7. **End confidently** — the last line should feel like a mic drop



Quick Cheat Sheet (Sticky Note / Teleprompter)

HOOK: "Imagine a call from your bank... it's a deepfake."
INTRO: "Meet VoxSentinalX – 10-layer forensic detection."
BENTO: Spectral → Breathing → Jitter → Neural LCNN → Fusion
SANDBOX: Wire Fraud (RED) → Authentic (GREEN)
LIVE: Start monitoring → Play deepfake → WATCH radar + gauge
→ "92% confidence" → Speak real voice → gauge drops
→ "Verified human"
FILE: Drag Elon Musk.wav → 3-second forensic timeline
TRAINING: "5,400 files, 86% accuracy, 11 engines, real-time"
CLOSE: "Trust needs to be verified. Thank you."



Recommended Recording Settings

SETTING	VALUE
Resolution	1920×1080 (1080p)
Frame Rate	60 fps
Format	MP4 (H.264)
Aspect Ratio	16:9
Audio	48 kHz stereo (mic + system audio)
Software	OBS Studio (free)
Bitrate	6000–8000 kbps

Script prepared for Smart India Hackathon (SIH 2026) — Problem Statement #26104 VoxSentinalX / All rights reserved